

$\Sigma x_2 := \text{Bool}, x_4 := \text{Nat} \rightarrow \text{dim } x_2 \Sigma x_2 := \text{Bool}, x_4 := \text{Nat}$

$\emptyset$

$\text{mgu} = \Sigma x_2 := \text{Bool}, x_4 := \text{Nat}$

① $\vdash (\lambda x : \text{Bool}. \text{zero}) \text{False} : \text{Nat}$

Note

but our mgu is in union it may produce no thing var it contains

10) $\text{W}(\text{if True then } (\lambda x : \text{zero}) \text{zero else } (\lambda x : \text{zero}) \text{false})$ ②

$\text{mgu} \Sigma \text{Bool} = \text{Bool}, \text{Nat} = \text{Nat} \rightarrow \emptyset$

③

$\text{if True then } (\lambda x : \text{zero}) \text{zero else } (\lambda x : \text{zero}) \text{false} : \text{Nat}$

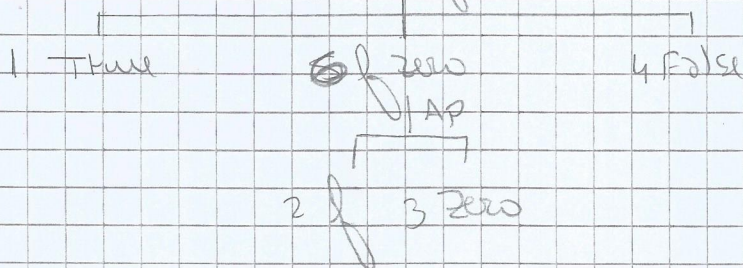
→ x and x in contextos diferentes, no les afecta el tipo de x .

④ $(\lambda f. \text{if True then } f \text{ zero else } f \text{ false}) (\lambda x. \text{zero})$ 10

9 $(\lambda f. \text{if True then } f \text{ zero else } f \text{ false})$ 7 $(\lambda x. \text{zero})$
| ABS | ABS

8 $\text{if True then } f \text{ zero else } f \text{ false}$

5 zero



① $\text{W}(\text{True}) \rightsquigarrow \vdash \text{True} : \text{Bool}$

② $\text{W}(f) \rightsquigarrow f : X_1 \vdash f : X_1$

③ $\text{W}(\text{zero}) \rightsquigarrow \vdash \text{zero} : \text{Nat}$

④ $\text{W}(\text{False}) \rightsquigarrow \vdash \text{False} : \text{Bool}$

⑤ $\text{W}(\text{zero}) \rightsquigarrow \vdash \text{zero} : \text{Nat}$